Aarnav Singh

+91-9971506703 | aarnavujji@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Security Operations Center Analyst at HCLTech with a B.Tech in Computer Science & Engineering (Cybersecurity). Delivers L1 security monitoring and alert triage, backed by hands-on experience in threat detection, log correlation, and IDS/IPS through a Zero-Trust adversarial-ML research paper (under review at Elsevier) and an Elasticsearch-based intrusion detection system. Fortinet NSE 1–3 and Google Cloud Cybersecurity certified, with strong networking foundations.

Experience

HCLTech – Security Operations Center Analyst (L1) October 2026 – Present
Senior Analyst (E1.2), Cybersecurity Operations *Noida, India*

- Member of the SOC team supporting a global financial-exchange client engagement, responsible for L1 security monitoring, alert triage, and incident escalation.
- Monitor and investigate alerts across enterprise security platforms including CrowdStrike (endpoint), Securonix (SIEM/UEBA), and Zscaler (secure web proxy).

HCLTech – Graduate Trainee August 2026 – October 2026
Graduate Trainee *Noida, India*

- Completing a structured pre-joining training program spanning cloud platforms (AWS, Azure, GCP, OCI), networking, Linux system administration, and DevOps/SRE practices (Terraform, Grafana, Prometheus, ITIL 4).
- Passed HCLTech's proctored pre-joining assessment, validating readiness across enterprise IT, cloud, and security fundamentals.

HCLTech – Intern May 2024 – July 2024
Cybersecurity, Networking & Cloud *Noida, India*

- Worked within enterprise-scale cybersecurity operations, building working knowledge of how security is structured and coordinated across a large IT organization, plus server environments and enterprise networking.
- Validated a Python automation script for provisioning Azure Virtual Machines, supporting cloud infrastructure automation.

Technical Skills

Security & Networking: Security Monitoring & Alert Triage — Threat Detection & Anomaly Analysis — Log Analysis & Correlation — Intrusion Detection (IDS/IPS) — Network Security Analysis — Vulnerability Assessment — Zero-Trust Architecture — Adversarial ML & Evasion Analysis — Networking (TCP/IP, DNS, Routing & Switching, VLANs/Subnetting) — Wireshark — Nmap — Nessus — Metasploit — Kali Linux — Cisco IOS

Cloud & DevOps: Amazon AWS — Microsoft Azure — Google Cloud Platform (GCP) — Terraform (IaC) — Docker — Prometheus — Grafana — Linux (RHEL) — Cloud Infrastructure Monitoring — ITIL 4 — SRE Practices

Programming & Tools: Crowdstrike Falcon — Securonix — Crowdstrike NG-SIEM — FortiGate — Python — PyTorch — SQL/MySQL — Elasticsearch — React.js — Git/GitHub — Jira — Windows — Technical Documentation & Research Writing

Education

SRM Institute of Science and Technology (SRMIST) India
B.Tech in Computer Science & Engineering, specialization in Cybersecurity *2022 – 2026*

Security Research & Projects

Adversarial Attack Detection in Zero-Trust Networks
GitHub: github.com/Aarnav-Singh/adversarial-ml-security-framework

- Built a six-layer Zero-Trust security system in Python/PyTorch, pairing ML-based intrusion detection with a context-aware policy engine.

- Designed a five-action policy engine (ALLOW / DENY / STEP UP AUTH / RATE LIMIT / ISOLATE) that cut evasion bypass rates from 65% to 2.2% without model retraining.
- Validated across 5 seeds on UNSW-NB15 and CICIDS-2017; co-authored the paper under review at the Journal of Information Security and Applications (Elsevier).

Hybrid Behavioral Intrusion Detection System
GitHub: github.com/Aarnav-Singh/Hybrid-Behavioral-Intrusion-Detection-System
- Built a full-stack IDS combining rule-based detection with Isolation Forest anomaly detection for behavioral threat analysis.
- Developed a real-time React dashboard (WebSockets, REST) for live threat visualization and alerting.
- Used Elasticsearch for log correlation and Prometheus for latency monitoring, containerized with Docker.

Certifications

Google Cloud Cybersecurity Certificate

Fortinet Certified in Cybersecurity: NSE 3 | NSE 2 | NSE 1

Cisco – Networking Basics

Cisco – Introduction to Cybersecurity

NPTEL – Advanced Computer Networks

Deloitte Australia (Forage) – Cyber Job Simulation